Internet of Things Course (420-521-VA)

# Final Project Phase 3

**Due Date:**

Thursday, November 13th , 2025


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# Overview

**Phase 3 of the IoT Smart Store project requires:**

1. Product management system with full CRUD operations
2. Barcode and RFID scanner integration for product identification
3. Self-checkout system with real-time cart updates
4. Customer account system with membership numbers and points
5. Point redemption system for discounts
6. Guest checkout with optional membership verification
7. Purchase history tracking and email receipt generation
8. Bilingual interface (English/French)
9. Inventory tracking with reception history

# System Design

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The system integrates barcode (USB) and RFID (CF600 UHF) scanners for product identification at self-checkout.

Customers can shop in three ways:

- Guest mode: Browse and purchase without an account
- Member login: Access points, purchase history, and account features
- Guest with membership: Enter membership number at checkout to earn points without logging in

The system connects to a Supabase database for product management, customer accounts, purchase tracking, and point accumulation. Admins can manage products, view customers, and monitor refrigerator temperatures from Phase 2.

## Data Flow

- Customers scan products using USB barcode scanner or RFID reader at /store/checkout
- Scanner inputs are invisible to prevent tampering - system auto-detects barcode vs RFID format
- Products appear in cart immediately with real-time price updates
- At checkout, customers can:
    - Redeem loyalty points (100 points = $1 discount)
    - Apply membership number for point earning
    - Complete purchase as guest, member, or new account
- System calculates points earned, sends email receipt, and updates purchase history
- Admin dashboard shows all customer data, purchase history, and product inventory

# Implementation

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- **Scanners**: USB barcode (plug & play), CF600 RFID with auto-detection
- **Database**: Supabase PostgreSQL with customer accounts, products, purchases, and inventory
- **Backend**: Flask with modular blueprints (auth, store, products, client, dashboard)
- **Frontend**: Bootstrap 5, vanilla JavaScript with invisible scanner inputs
- **Hardware**: Phase 2 ESP32 sensors and Raspberry Pi GPIO still functional
- **Internationalization**: Flask-Babel with English/French translation support
- **Point System**: Customers earn points on purchases, redeem in multiples of 100

- Repository: https://github.com/Crisguts/IoT-Semester-Project-

# Task Distribution

| Team Member | Task |
| --- | --- |
| Cristian | Refactored project to cleaner folder structure with modular blueprints. Implemented complete self-checkout system with invisible scanner inputs, point redemption, and guest checkout flow. Built cart management API and purchase completion logic. Integrated barcode and RFID scanning with auto-detection. Created email receipt system. Developed point calculation and redemption system. Fixed Phase 2 features. |
| Trevor | Organized all templates, JavaScript, and CSS files into clean structure. Designed the application UI/UX with Bootstrap 5 components. Created responsive layouts for store, cart, checkout, and account pages. Styled dashboard gauges and product management interfaces. Implemented consistent design language across all pages. Fixed Phase 2 features. |
| Parth | Researched and partly implemented RFID functionality including CF600 reader integration. Researched barcode scanner functionality. Proposed and implemented guest checkout feature on Cristian's laptop. Contributed to RFID scanner code and testing. Hosted new Supabase database model for refactored project with customer accounts and purchase tracking. Fixed Phase 2 features. |
| Adlane | Researched and partly provided code for RFID functionality. |